

MIAE 215 Objectives

↕ MECH

Learn how to develop C++ programs and algorithms

How to solve engineering problems using C++

How to effectively use a C++ compiler → offline

Introduction to microcontroller (Arduino) programming

Why Learn Computer Programming ?

Very useful skill in industry (software development)

Very useful for engineering (analysis, automation)

Programming is the language of automation

Many interesting well paying jobs in Mech / Indu / Aero Engineering require good programming skills

Important MIAE Applications

Numerical analysis (finite element method, etc.)

Simulation (dynamic systems, virtual reality)

Optimization (design, scheduling, etc.)

Automation (manufacturing, smart machines, etc.)

Robotics and control systems (automotive, aerospace, manufacturing, machine learning, etc.)

Instrumentation and measurement (experiments, etc.)

Why Learn C++ ?

C++ is the fastest structured (ie organized) general purpose programming language

C++ is the best language for applications that need to be fast and/or have direct hardware access such as:

other languages are easier to learn, if you learn C++

- mechatronics / robotics other languages are easy
- instrumentation and measurement
- Internet of things (IoT) many other languages have similar syntax (Java, Javascript, etc.)
- simulation
- video games / virtual reality
- operating systems, compilers, etc.

Basics of Computing

Computer

A digital electronic device for computation composed of:
CPU (processors), memory (RAM, disk, etc.),
Input / Output (IO)

Computer Program

A sequence of instructions for the computer.
Often referred to as software.

Operating System (OS)

Software that manages all other programs and
provides services (disk, graphics, etc.) for them

→ Windows, OSX, Linux
Unix

Programming Language

A system that allows humans to provide written instructions to the computer.

These instructions can then be translated to a form the computer can directly use (ie machine language -- 1s and 0s)

01110110110 $\Rightarrow$ computer

① Compiler

Software that translates programs written in C, C++, Fortran, etc. into machine language -- all at once

$\Rightarrow$ do something
execute program

Some languages (Python, etc.) translate one instruction at a time using Interpreter software -- slow but flexible

② $\downarrow$ MATLAB

Phases of C++ Program Development:

1. Edit (a text file)

2. Preprocess

3. Compile

4. Link

5. Execute

6. Test, Debug, Optimize

Build

(remove errors/bugs)

object files

Source files (code)

